IoT based wearable device for assisting Autism Spectrum Disorder (ASD) patients

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***Abstract*—Autism Spectrum Disorder (ASD) is a neurological disorder which is caused by imbalance in neuronal activities. This affects the communication, behavior pattern, and social life of people diagnosed with ASD. Autistic patients suffer from multiple health related issues such as sensory processing disorder, anxiety, seizure, sleep disorder and gut related issues. People living with autism face lot of difficulty to live a better life. So, there is a need to develop a technology based assistive device to improve the quality of life for ASD patients and their family members. Recent advances in Internet of Things (IoT) and wearable technologies offer a promising way to support individuals with autism in real-time. This paper proposes an IoT-based wearable support system designed to track, guide, and assist individuals with ASD in their daily lives. The proposed device integrates multiple sensors to track parameters like heart rate, Stereotypical Motor Movement (SMM) like hand flapping motion and ASD patient location. The collected data is transmitted to caregivers and therapists through cloud platforms and mobile apps, enabling timely alerts during stress or abnormal behavior. The system also enables the therapist to prepare a treatment plan by analyzing the behavior pattern of the patient. Features such as music therapy, emotion detection, and personalized feedback aim to calm and engage the child. The proposed assistive system is implemented with hardware prototype using ESP 32 microcontroller and multiple sensors such as pulse oximeter, inertial measurement unit (IMU) and GPS tracker to justify its functionality and feasibility. Experimental evaluation shows improved communication, anxiety management, safety and efficient treatment plan in ASD patient using the device. By sending alerts in real time through mobile connectivity the proposed device enhances the quality of life for autistic individuals. This paper contributes to the growing field of assistive technologies for neurodevelopmental disorders.**

Keywords—ASD, SMM, hand flapping, IoT, IMU sensors, pulse oximeter

I. Introduction

Autism Spectrum Disorder (ASD), typically known as Autism, is a neurodevelopmental disorder which is characterized by lack of communication, difficulties in social interaction, limited and repetitive interests, irregularities in behaviour pattern, and sensitivity to sensory stimuli [1]. It is mainly caused by imbalance in excitation and inhibition of neurons in the nervous system [2]. Autism affects functioning of the brain of individuals diagnosed with ASD. Symptoms of Autism usually appear early in childhood and can differ greatly from person to person. These differences often depend on the individual's thinking and language skills, as well as the presence of other physical health conditions. This is why the term "Autism" was changed to "Autism Spectrum Disorder". It is observed that autistic people face other health related challenges such as seizure disorder, sensory processing disorder, gut issues, sleep disturbances, anxiety and phobias, attention deficit and hyperactivity disorder [3]. These challenges make the life of people with autism and their family members very difficult and helpless. Living with Autism Spectrum Disorder (ASD) often means facing sudden bouts of anxiety or sensory overload that can disrupt daily routines and social interactions. This fact indicates a need to develop a portable and compact device to assist people with ASD to live a better and safe life.

Many individuals with ASD use repetitive behaviours—such as hand flapping, rocking or hitting themself—to self-soothe, but these actions can hamper learning and also can create safety risks. Sometimes, the lack of communication, sensory processing disorder and other difficulties in ASD people leads to increased heart rate due to anxiety and restlessness. In addition, wandering outside safe areas poses a constant concern for caregivers, who cannot always provide uninterrupted supervision. These challenges underscore the need for a reliable, always-on support system that can detect signs of distress and notify caregivers in real time.

In recent years the technological development of assistive devices such as Augmentative and Alternative Communications (AAC) solutions has benefitted people with ASD to overcome their day-to-day life difficulties [4]. However, Internet of Things (IoT) based wearable devices like E4 wristbands enable seizure monitoring for individuals with autism. We went through a lot of literature to find the work done in the field of assistance for ASD patients. Some of the observations are mentioned in Table 1. It is observed that today's wearable device for ASD has monitored and reported about only a single parameter like heart rate [5-8], stereo typical movements [9-10] and safety issues [11-13]. The authors in [14-16] have proposed complete theoretical analysis of the support system for autism but could not built a hardware prototype to implement the same. To truly support patients, we need a next-generation system that not only detects problems like rising heart rate, repetitive hand movements, or wandering beyond safe zones but also guide users with personalized behaviour analysis which can help the therapist to prepare treatment plan. Such a device must collect and store data from multiple sensors, adjust its warning levels for each individual, and fit into a small, low-power device.

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Table 1 Comparative Analysis of existing IoT based assistive device

Reference Paper No.	Heart Rate Measurement	Stereotypic Movement Measurement	GPS/Location Tracking	Prototype Built	Cloud Storage Possibility	SMS Alert Possibility
[5]	Yes	No	No	No	No	No
[6]	Yes	No	No	No	No	No
[7]	Yes	No	No	Yes	Yes	Yes
[8]	Yes	No	No	No	No	No
[9]	No	Yes	No	Yes	Yes	No
[10]	No	Yes	No	Yes	Yes	No
[11]	No	No	Yes	No	No	No
[12]	No	No	Yes	Yes	Yes	Yes
[13]	No	No	Yes	Yes	No	Yes
[14]	Yes	No	Yes	No	Yes	Yes
[15]	No	No	No	No	No	No
[16]	No	No	No	No	No	No
Proposed System	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**

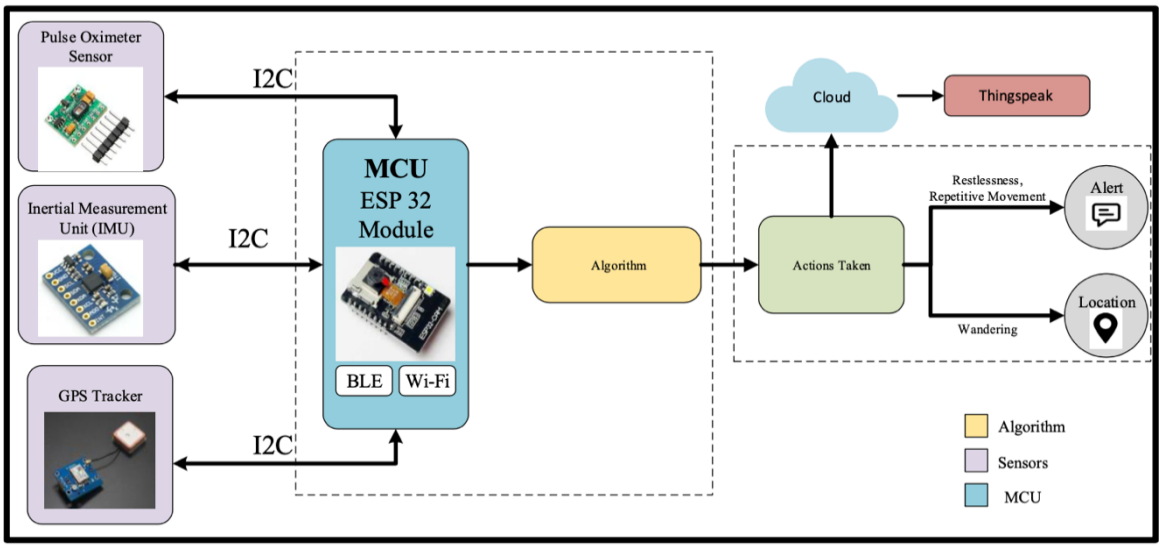


Figure 1 – Block diagram of proposed IOT based assistive wearable device for ASD pateints

In this work we proposed a compact IoT platform that directly connects a pulse oximeter, a motion sensor, and a Global Positioning System (GPS) module to an ESP32 microcontroller and hence we can track heart rate, movement patterns, and location in real time and send customized alerts to caregivers as well as parents. The proposed IoT system also store the collected data at cloud to analyse the behaviour pattern of the ASD patient by the concerned therapist. The proposed compact IoT-based assistance platform is tailored for individuals with ASD. The device first applies simple threshold checks—such as elevated heart rate or repeated, high-variance movements—to identify potentially meaningful events. It then uses a small, efficient classification model to place the wearer's state into one of four categories: calm, restless, wandering, or urgent. When an alert is triggered, a concise notification is sent via Wi-Fi or Bluetooth to a caregiver's smartphone, and key data are logged to the cloud for later analysis. This approach delivers fast, context-aware alerts while protecting user privacy and ensuring low power consumption, helping families manage daily life with greater confidence and peace of mind.

II. Proposed System

In this work, we developed an IoT-based system that continuously monitors the physical and emotional state of individuals with Autism Spectrum Disorder (ASD). The goal is to detect signs of stress, restlessness, or unsafe movement and to alert caregivers in real-time. The system is designed to be wearable, lightweight, and wireless, allowing free movement while ensuring close monitoring as shown in Figure 1.The core of our assistance system is an ESP32 microcontroller, chosen for its built-in Bluetooth Low Energy and Wi-Fi radios, plus its ability to run real-

time code right on the device. Three smart sensors—an optical pulse oximeter, a six-axis inertial measurement unit (IMU), and a GPS tracker—are each connected to the ESP32 over a serial Inter Integrated Circuit (I2C) interface. The pulse oximeter continuously measures the wearer's blood-oxygen saturation and heart rate, providing vital signs that can signal stress or anxiety. At the same time, the IMU reports acceleration and rotational motion along three axes. An IMU sensor can detect hand flapping by measuring sudden or repetitive movements of the hand or arm. It tracks changes in speed and direction using built-in accelerometers and gyroscopes. This helps in identifying patterns of stereotypic behavior common in individuals with autism. The data can then be used to monitor and respond to such behavior in real time. Meanwhile, the GPS unit keeps track of the wearer's real-time location and thus the system always knows where the person is moving.

All of these data streams flow into ESP32 where the lightweight on-device algorithm constantly analyzes the patterns it sees. If it recognizes signs of restlessness—like a sudden spike in heart rate paired with repetitive motion—the system immediately triggers an alert and sends notification over Bluetooth Low Energy or Wi-Fi to a paired caregiver's smartphone. If the patient strays outside a predefined safe zone the software automatically flags the GPS coordinates and pushes the location to the caregiver which helps them to respond quickly. At the same time if the ESP32 is still connected to Wi-Fi then it builds a ThingSpeak update URL containing different fields. An HTTP GET to ThingSpeak logs these values in the user's channel where they can be graphed or analyzed later. This information can be later used by therapists for future examination and treatments.

Some remedies which can be taken by parents/caretakers for the patients are:

- Set up a calm, quiet spot with soft lighting and a few favorite items to help them feel safe.
- Give gentle, comforting pressure—like wrapping in a light blanket or having them push against a wall—to help their body relax.
- Offer easy-to-use fidget tools or noise-cancelling headphones to distract and soothe their senses.
- Keep a consistent daily routine with picture charts or simple schedules.
- If the patient feel too restless or anxious, guide them through a few deep breaths or let them move briefly—like swinging gently or doing a few hops—to let off steam.

The ESP32 takes in data from the pulse oximeter, motion sensor, and GPS. First, it does a quick pre-check and if the heart rate goes over 105 bpm the movement gets too fast or in case the GPS leaves a radius of 100 m from the reference point, it gives alert to the caretaker using telegram messaging. Every other data is discarded out from the lot. Next, those flagged moments go through a small and quick classifier on the ESP32. It looks at the hand movement and heart-rate spikes together. Then it sorts the wearer's state into one of four simple categories according to the movement and heart-rate. As soon as it decides a clear alert is sent over Wi-Fi or Bluetooth to the caregiver. The notification helps to take appropriate action for ASD patients.

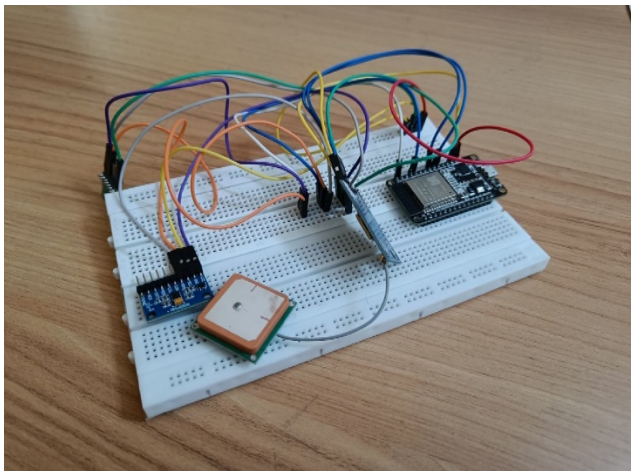


Figure 2 – Hardware setup of proposed IoT based assistive system

III. Implemenation

The implementation of the IoT based prototype system consists of both hardware and software component. It is a 3-tier structure which includes (1) collecting data from different sensors, (2) analysing and storing data, and, (3) sending the data to selected recipients. The hardware set up for the proposed IoT based system is shown in figure-2. The ESP-32 microcontroller is interfaced with sensors to establish a real-word monitoring environment.

The complete specification of ESP 32 includes a dual-core 32-bit processor, Wi-Fi, and Bluetooth along with a wide array of peripheral interfaces. ESP32 is powered via micro universal serial bus (USB) port or Vin from external supply. The three sensors pulse oximeter, IMU and location device are connected to ESP 32 through I2C protocol and operates around 3.3V to 5V power supply. To supply power to all the sensors, we connected the 3.3V pin of the ESP32 to the VCC pin of each sensor — the MAX30102, MPU-6500, and NEO-6M. Then, we connected the GND pins of all these sensors to one of the ground pins on the ESP32. This setup allows all the sensors to use the same power source and turn on properly without needing any voltage converter. All sensors are connected to the ESP32 in the following manner. Both the heart rate and motion sensors share the same I2C bus such as GPIO (General Purpose Input/Output) 21 on the ESP32 goes to the Serial Data (SDA) pin of each sensor module, and GPIO 22 goes to the Serial Clock (SCL) pins. Because the sensor breakouts include pull up resistors and thus no additional resistors are required. The GPS module uses a separate serial connection where its transmitter (TX) pin is wired to ESP32 Receiver (RX2) pin (GPIO 16). Similarly, its RX pin is connected to ESP32 TX2 (GPIO 17) pin with communication set to 9600 baud. By connecting power, I2C and Universal Asynchronous Receiver Transmitter (UART) in this manner, the ESP32 can continuously read heart-rate, motion, and location data in a single, straightforward loop.

The ESP32 firmware begins by including libraries for MAX30105 pulse oximeter, MPU-6500 IMU, and Tiny GPS Plus module. To create the reference and threshold levels, we referred a standard value of heart rate in NORMAL and ANXIETY conditions through the information given in Table 2. A pulse oximeter sensor (MAX30102) consists of a light emitting diode (LED) pair such as one red & another infrared and a photodetector

Table 2 – Heart rate differentiated on the basis of gender and age with ASD during anxiety [6][7][8]

Age Group	Boys With ASD & Anxiety	Girls With ASD & Anxiety
5–7 years	95–110 bpm	97–112 bpm
8–10 years	90–105 bpm	92–108 bpm
11–13 years	85–100 bpm	87–102 bpm
14–15 years	80–95 bpm	82–98 bpm

Table 3 – Acceleration values of regular movements [9,10]

Cases	Readings
Space	0
Forced not to move	≈ 1.1
Under fan wind	≈ 1.6
Ordinary hand movements	≈ 2.0 – 2.6
Fast movements	≈ 3.0 – 5.0
Very fast movements	≥ 6.0
Free fall	9.8

Table 4 – Classification of wearer's state

Situation	Status	Alert
Normal heart rate + Ordinary hand movements	Normal	❌
Normal heart rate + Fast hand movements	Restlessness	✅
Heart rate above normal + Ordinary hand movements	Anxiety	✅
Heart rate above normal + Fast hand movements	Anxiety + Restlessness	✅

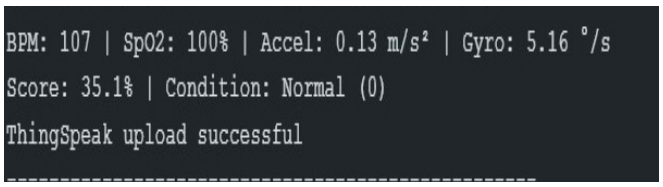
placed on opposite sides of a thin body part (such as a fingertip or earlobe). As arterial blood pulses through tissue and the variations in absorption of two wavelengths are measured by a photodetector. By comparing how much amount of red and infrared light are absorbed with each heartbeat, the sensor calculates the percentage of oxygenated haemoglobin (SpO_2) and also derives the pulse rate in real time. An Inertial Measurement Unit (MPU6500) is an electronic sensor module that combines accelerometers, gyroscopes, and sometimes magnetometers to measure linear acceleration, angular velocity, and orientation. By fusing these measurements an IMU can track motion and determine the device's position and attitude in real time. The code updates accelerometer and gyroscope readings. Acceleration in all three axes is converted from "g" units to meters per second squared and then combined into one magnitude small fluctuations under ±0.5 g in the gyro. We referred standard values of accelerations for different movements through the information given in Table 3.The NEO-6M is a GPS receiver module which is used to determine geographic location (latitude, longitude, altitude) and time by decoding signals from GNSS satellites. It is commonly employed for navigation, tracking, and geofencing in embedded and IoT devices.

For programming various hardware we have chosen python programming language on Arduino IDE interface. In setup routine the device opens a serial console which connects to wi-fi and initializes each sensor. In its main loop the code reads 100 samples from MAX30102 to calculate heart rate and SpO_2, converts raw accelerometer and gyroscope data into combined motion vectors, and parses GPS sentences to track distance from an origin point. Simple threshold rules then classify movement intensity into four states such as normal, anxiety, restlessness, or urgent and thus trigger a telegram alert if needed. Simultaneously in every second the ESP32 logs heart rate, SpO_2 and motion vectors. If either (a) the condition code is non-zero or (b) the distance from origin exceeds 100 m and at least 30s have passed since the last alert, the device composes a text message through Telegram and geo alerts. This message is URL encoded and sent via an HTTP GET request to the Telegram Bot API. We record the time of each alert so we don't spam notifications. There are condition codes sent to ThingSpeak via HTTP which ensures that caregivers receive timely notifications and data is stored at cloud for later analysis. This model further can be scaled by developing a mobile application, making information more easily accessible to the caregivers.

IV. Results

In this work we have implemented IoT based smart assistive environment prototype to support individuals with ASD. This wearable system is designed in such a way that can be wear or carried by ASD patient to enhance and support their daily life. The functionality of the proposed IoT based system is tested and validated with real time sensor data and monitored for different set of cases which are discussed below. Table 4 shows how the proposed system classifies the wearer's state based on their heart rate and hand movement. If either the heart rate is high or hand movement is fast an alert is triggered to indicate possible anxiety or restlessness.

To evaluate and validate the feasibility of the proposed IoT based system we monitored different set of cases and noted different results as shown in figure 3 and 4.



a. Output for NORMAL condition

```
BPM: 48 | SpO2: 89% | Accel: 5.62 m/s² | Gyro: 191.36 °/s
Score: 55.6% | Condition: Restlessness (1)
Telegram alert sent
ThingSpeak upload successful
```

b. Output for RESTLESSNESS condition

```
BPM: 250 | SpO2: 100% | Accel: 0.10 m/s² | Gyro: 3.74 °/s
Score: 40.6% | Condition: Anxiety (2)
ThingSpeak upload successful
```

c. Output for ANXIETY condition

```
BPM: 136 | SpO2: 100% | Accel: 5.37 m/s² | Gyro: 208.62 °/s
Score: 71.2% | Condition: Anxiety + Restlessness (3)
ThingSpeak upload successful
```

d. Output for ANXIETY + RESTLESSNESS condition

Figure 3 Outputs for different condition of ASD patient

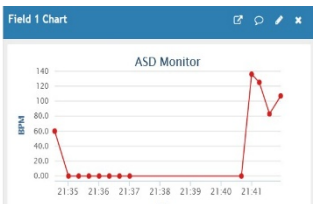
a. Telegram Alert for RESTLESSNESS condition

b. Telegram Alert for ANXIETY condition

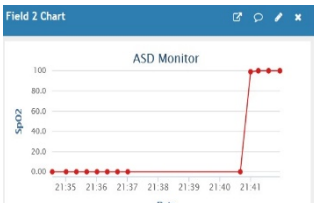
c. Telegram Alert for ANXIETY + RESTLESSNESS condition

Figure 4 – Telegram Alerts

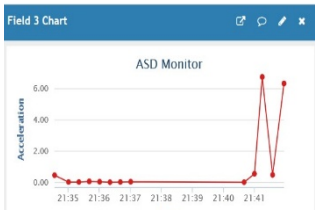
Figure 3 shows the output condition measured in different real-time situations as mentioned in table 4. Heart rate and hand movements of an autism patient is measured and weighted averagely to find the condition of the wearer. Figure 4 shows the telegram alert for the three conditions indicated in table 4, other than NORMAL, which will alert the caretakers timely about any discomfort faced by the patients.



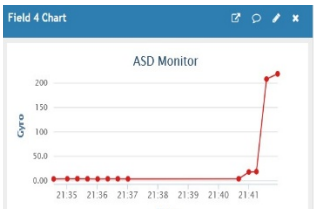
(a) BPM v/s Time



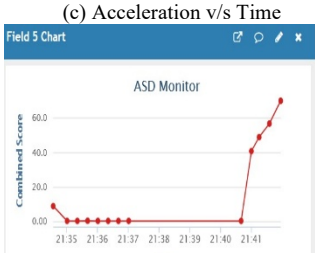
(b) SpO_2 v/s Time



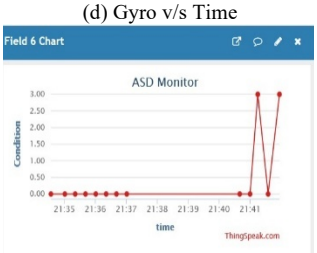
(c) Acceleration v/s Time



(d) Gyro v/s Time



(e) Combined Score v/s Time



(f) Condition v/s Time

Figure 5 Cloud logging (ThinkSpeak) in ESP32 Sketch

The extracted motion graphs in Figure 5 represents repetitive motion behaviour, BPM, SpO2 and combined score of these quantities that can help the therapists as well as clinicians to learn about real time movements and heart rate of the individuals with ASD through a cloud-based platform. Based on the data, therapists and clinicians can study the graphs and different patterns that follow. This will help them to give specific therapy to help the autistic individuals get out of their repetitive motion behaviour. The innovation proposed in this paper will also help the caretakers to help the autistic individuals when needed.

V. Conclusion

In this work we presented an IoT-based wearable solution designed to assist individuals with Autism Spectrum Disorder (ASD). The system enables continuous monitoring and tracking of heart rate, behavioral parameters such as hand flapping and location of the wearer through integrated sensors. Real-time data transmission and alerts empower caregivers as well parents to respond promptly during distress or abnormal behavior. Our findings indicate that such wearable devices can significantly enhance quality of life, independence, and safety for individuals with ASD. The data stored in cloud helps the therapist to analyze the behavior of the patient and provide necessary therapies to them. By combining technologies such as IoT and cloud computing the device adapts to user needs and provides personalized feedback. This research paves the way for developing more accessible, intelligent, and supportive technologies for neurodivergent individuals.

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